ENV S 193DS Final

Aylin Orozco

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GitHub Repository: here

```r
# Set Up
library(tidyverse)
library(here)
library(janitor)
library(knitr)
library(DHARMa)
library(ggeffects)
library(ggplot2)
library(effsize)

nests <- read_csv(here("data/nest_data_final.csv"))

my_data <- read_csv(here("data/my_data - Sheet1.csv"))
```

1 Problem 1. Research writing

a. In Part 1, my co-worker used a logistic regression model to test whether distance to water edge predicts the probability of American Avocet microhabitat use. In Part 2, they used a chi-square test of independence to evaluate whether bird species (American Avocet, Forster's Tern, Black-necked Stilt) are associated with different habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat).

b. American Avocets are more likely to use habitat closer to the water's edge, suggesting that proximity to water is an important factor shaping where they forage and rest. This pattern indicates that shoreline accessibility likely influences habitat selection.

Bird species differ in how they use available habitats across the San Francisco Bay. Each species tends to be associated with certain habitat types, indicating that habitat structure and conditions likely shape where different shorebirds are found.

c. One additional variable that could influence American Avocet habitat use is water depth (cm), measured at each microhabitat sampling point using a graduated meter stick during low tide surveys. Water depth likely affects avocet foraging efficiency because they feed in shallow water where prey is more accessible and wading is energetically efficient.

2 Problem 2. Data analysis

a. Clean and Wrangle

```r
nests_clean <- nests |>
  select(height_cm, ant_species, case_control) |>
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  mutate(
    ant_species = factor(
      ant_species,
      levels = c("AB", "RRB", "BBR")
    ),
    ant_full = case_when(
      ant_species == "AB" ~ "C. sjostedti",
      ant_species == "RRB" ~ "C. mimosae",
      ant_species == "BBR" ~ "C. nigriceps"
    )
  )

str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ ant_species : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_full    : chr [1:270] "C. mimosae" "C. mimosae" "C. mimosae" "C. mimosae" ...
```

```r
sample_n(nests_clean, 10)
```

```
# A tibble: 10 x 4
   height_cm ant_species case_control ant_full
       <dbl> <fct>              <dbl> <chr>
 1      114  RRB                    0 C. mimosae
 2      247  RRB                    1 C. mimosae
 3      131  BBR                    0 C. nigriceps
 4      194  RRB                    0 C. mimosae
 5      238. RRB                    1 C. mimosae
 6      310  RRB                    0 C. mimosae
 7      420. RRB                    1 C. mimosae
 8      201. RRB                    1 C. mimosae
 9      129  RRB                    0 C. mimosae
10      110. RRB                    0 C. mimosae
```

b. In this dataset, a value of 1 indicates that a tree contained a bird nest, meaning it was actively selected for nesting by birds. A value of 0 indicates that the tree was an available but unused site, meaning no nest was present in that tree during sampling.

c. Tree height (height_cm) is a continuous variable (measured in cm). As tree height increases, the probability of nest occurrence would likely increase because taller trees may provide greater visibility, reduced predation risk, and more suitable structural support for nests. This relationship is supported in the Methods section.

Ant species (ant_species) is a categorical variable with three levels (AB, RRB, BBR) representing different acacia-ant mutualists. As ant species changes, the probability of nest occurrence may vary because different ant species differ in aggressiveness and defensive behavior, which can either deter predators (making nesting safer) or deter birds from nesting in highly aggressive trees. This is described in the Introduction and Methods sections.

d.

```r
models <- data.frame(
  `Model number` = c(1, 2),
  `Model description` = c(
    "case_control ~ height_cm + ant_species (no interaction)",
```

```r
    "case_control ~ height_cm * ant_species (includes interaction)"
  ),
  `Response variable` = c("case_control", "case_control"),
  `Interaction term` = c("No", "Yes")
)

knitr::kable(models)
```

Model.number	Model.description	Response.variable	Interaction.term
1	case_control ~ height_cm + ant_species (no interaction)	case_control	No
2	case_control ~ height_cm * ant_species (includes interaction)	case_control	Yes

e.

```r
# Model 1: additive logistic model (no interaction)
model1 <- glm(case_control ~ height_cm +
                ant_species,
              data = nests_clean,
              family = binomial)

# Model 2: interaction model
model2 <- glm(case_control ~ height_cm * ant_species,
              data = nests_clean,
              family = binomial)
```

f.

```r
# AIC comparison used for model selection
AIC(model1, model2)
```

```
       df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model as determined by Akaike's Information Criterion (AIC) was the interaction model predicting case_control (probability of nest occurrence) as a function of height_cm, ant_species, and their interaction. This model had the lowest AIC value (237.80), indicating that allowing the effect of tree height to vary among ant species improved model fit compared to the additive model.

g.

```
# simulate residuals from the best model
sim_res <- simulateResiduals(model2)

# # DHARMa diagnostics used to assess fit
plot(sim_res)
```

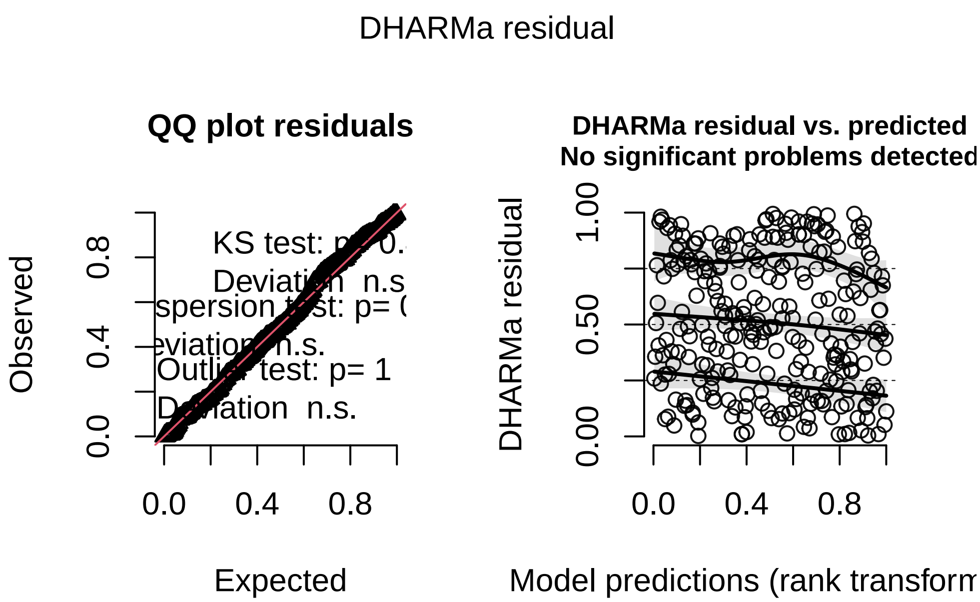


Figure 1. The DHARMa diagnostic plots were used to assess model fit for the selected interaction model. Even though there is some visual crowding present due to the large number of observations and binary response structure, the residual patterns were evaluated for systematic deviations from uniformity to determine whether the model appropriately captures variation in nest occurrence.

h.

```
# install if needed
# install.packages("ggeffects")

# Generate model predictions
```

```r
pred <- ggpredict(model2, terms = c("height_cm", "ant_species"))

# Plot predictions + underlying data
ggplot() +

  # underlying data
  geom_jitter(
    data = nests_clean,
    aes(x = height_cm, y = case_control),
    width = 0,
    height = 0.03,
    alpha = 0.3,
    color = "darkgray"
  ) +

  # 95% confidence interval
  geom_ribbon(
    data = pred,
    aes(x = x, ymin = conf.low, ymax = conf.high),
    fill = "lightblue",
    alpha = 0.4
  ) +

  # prediction line
  geom_line(
    data = pred,
    aes(x = x, y = predicted),
    color = "orchid4",
    linewidth = 1
  ) +

  # separate panel for each ant species
  facet_wrap(~group) +

   # label axes
  labs(
    x = "Tree height (cm)",
    y = "Probability of nest occurrence"
  ) +

  # remove gray background and gridlines
  theme_classic() +
```

```r
  # remove legend
  theme(legend.position = "none")
```

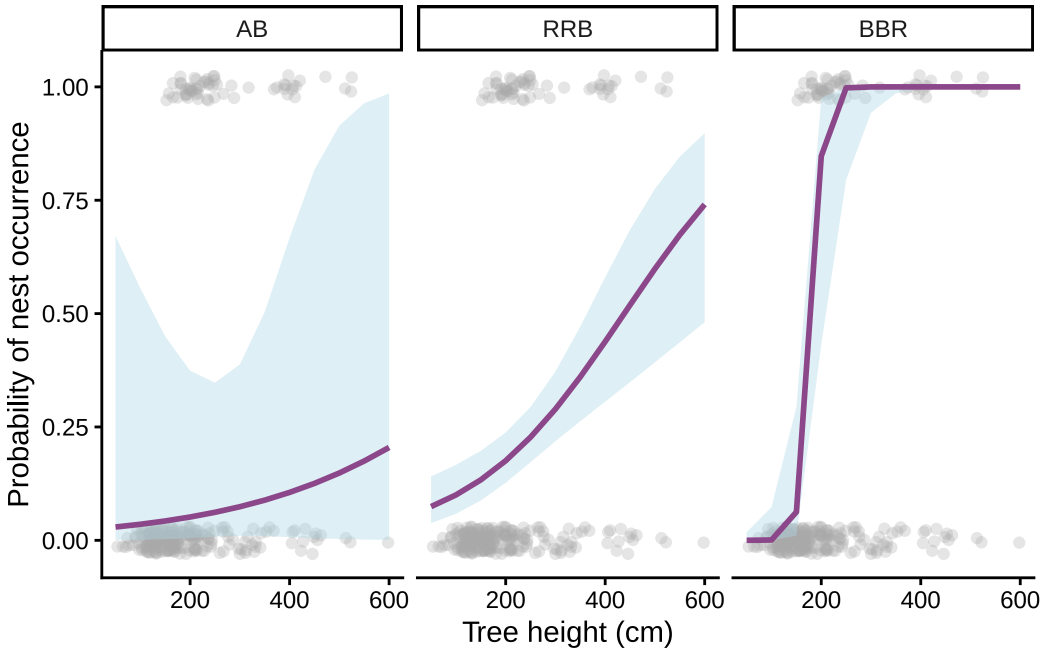


Figure 2 Predicted probability of bird nest occurrence by tree height and acacia ant species. Model-predicted probabilities of nest occurrence are shown as a function of tree height (cm) for trees inhabited by Crematogaster sjostedti, Crematogaster mimosae, and Crematogaster nigriceps. Solid lines represent predictions from the selected logistic regression model, shaded ribbons represent 95% confidence intervals, and points show the observed nest occurrence data (1 = nest present, 0 = no nest present). Data source: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

j.

```r
# New data for tallest tree in dataset
newdata <- data.frame(
  height_cm = 600,
  ant_species = factor(
    c("AB", "RRB", "BBR"),
    levels = levels(nests_clean$ant_species)
  )
)
```

```
# Predicted probabilities
predictions <- predict(
  model2,
  newdata = newdata,
  type = "response"
)

# Combine species and predictions
prediction_table <- data.frame(
  ant_species = newdata$ant_species,
  predicted_probability = predictions
)

prediction_table
```

```
  ant_species predicted_probability
1          AB             0.2048017
2         RRB             0.7407857
3         BBR             1.0000000
```

k. Figure 2 shows that the probability of nest occurrence was influenced by both tree height and acacia ant species, and that the effect of tree height differed among ant species. At the tallest observed tree height (600 cm), the predicted probability of nest occurrence was 0.20 for trees occupied by *C. sjostedti* (AB), 0.74 for trees occupied by *C. mimosae* (RRB), and 1.00 for trees occupied by *C. nigriceps* (BBR). These results indicate that taller trees were generally more likely to contain nests, but the strength of this relationship depended on the species of acacia ant inhabiting the tree. The observed pattern is consistent with the findings of Lujan et al. (2023), which suggest that birds preferentially select trees occupied by certain ant species because the ants aggressively defend their host trees, potentially reducing disturbance from browsing mammals and creating safer nesting sites.

3 Problem 3. Affective and exploratory visualizations

a. For Homework 2, I focused on showing the data clearly and analyzing it with standard graphs. The boxplot compares walk times between Reeboks and Vans, and the scatterplot looks at whether temperature affects trip duration. For Homework 3, I took the same data and turned it into a game-style visualization called Shoe Grand Prix, where the data is presented as a race from home to work with rankings, stats, and visual storytelling.

All of the visualizations use the same walking dataset and are trying to explain what factors affect my walk to work. They all include information about trip duration and compare different variables, even though they present the information in different ways. The main goal of all three is to help viewers understand patterns in the data.

The boxplot shows that my walks wearing Reeboks were usually shorter than my walks wearing Vans. The scatterplot shows that temperature doesn't seem to have much of an effect on how long my walks took. The affective visualization highlights the same result by showing Reeboks as the winner of the race. The patterns aren't really different between the visualizations because they're based on the same data, but the affective visualization makes the comparison more fun and easier to connect with.

One suggestion I got was to make the visualization feel more personal and tell more of a story instead of just showing the data. I tried to do that by turning my walk into a race course and adding details like hills, puddles, weather, and shoe "stats." I also kept the boxplot in the final design so the visualization still includes the actual data while being more engaging and creative.

b. One analysis I could use for this dataset is a two-sample t-test to compare walking duration between Reebok and Vans shoes. This analysis is appropriate because my response variable, walking duration (in seconds), is a continuous variable, and my predictor variable, shoe type, is a categorical variable with two groups (Reebok and Vans).

A two-sample t-test would help answer my research question because it tests whether the average walking duration differs between the two shoe types. Since I am interested in comparing the mean walk times of two independent groups, a two-sample t-test is a good fit for the data and the question I am asking.

c.

3.1 Assumption Checks

```r
# Check normality for each shoe type
my_data |>
  group_by(`Shoe Type`) |>
  summarize(
    p_value = shapiro.test(`Duration (s)`)$p.value)
```

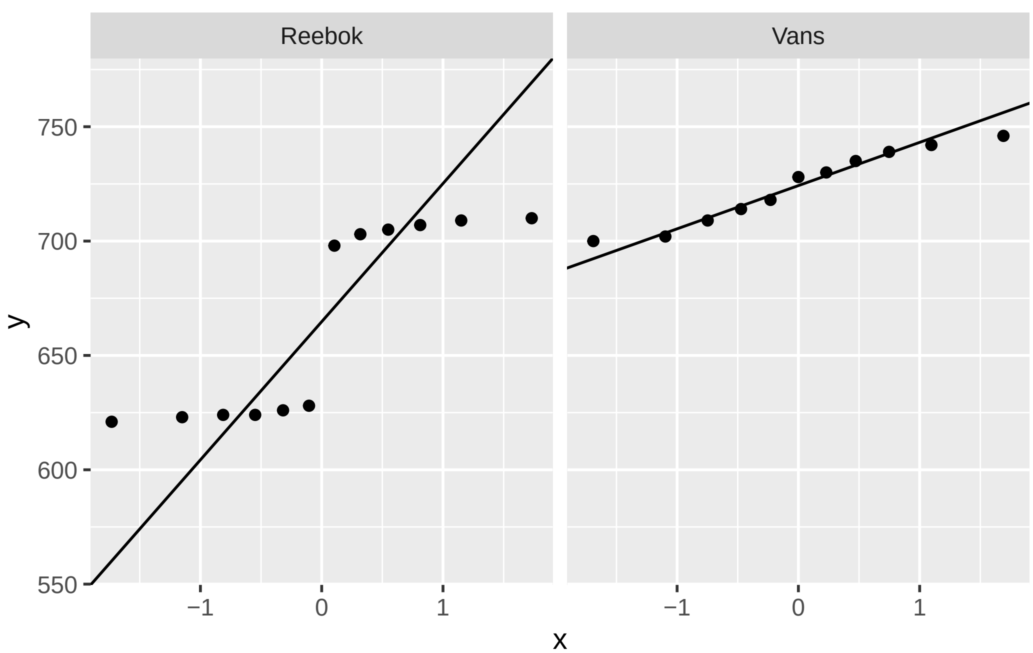
```
# A tibble: 2 x 2
  `Shoe Type` p_value
  <chr>         <dbl>
1 Reebok      0.00114
2 Vans        0.501
```

```r
# Check equal variances
var.test(`Duration (s)` ~ `Shoe Type`,
         data = my_data)
```

```
    F test to compare two variances

data:  Duration (s) by Shoe Type
F = 6.8555, num df = 11, denom df = 10, p-value = 0.005015
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
  1.870577 24.170259
sample estimates:
ratio of variances
          6.855505
```

```r
# Q-Q plots for visual assessment of normality
ggplot(my_data, aes(sample = `Duration (s)`)) +
  stat_qq() +
  stat_qq_line() +
  facet_wrap(~ `Shoe Type`)
```



3.2 Two Sample t-Test

```
# running t-test
t.test(`Duration (s)` ~ `Shoe Type`,
       data = my_data,
       var.equal = FALSE)
```

```

    Welch Two Sample t-test

data:  Duration (s) by Shoe Type
t = -4.4791, df = 14.379, p-value = 0.0004876
alternative hypothesis: true difference in means between group Reebok and group Vans is not e
95 percent confidence interval:
 -87.29395 -30.85757
sample estimates:
mean in group Reebok   mean in group Vans
            664.8333             723.9091
```

3.3 Calculate Effect Size

```
# calculate effect size
cohen.d(`Duration (s)` ~ `Shoe Type`,
        data = my_data)
```

```

Cohen's d

d estimate: -1.807343 (large)
95 percent confidence interval:
     lower      upper
-2.8372312 -0.7774555
```

d.

```
ggplot(my_data,
       aes(x = `Shoe Type`,
           y = `Duration (s)`,
```

```r
           fill = `Shoe Type`)) +

  # Create boxplots
  geom_boxplot(alpha = 0.7,
               outlier.shape = NA) +

  # Underlying data
  geom_jitter(width = 0.15,
              alpha = 0.7,
              size = 2) +
# label axes
  labs(
    x = "Shoe Type",
    y = "Walking Duration (seconds)"
  ) +
# change colors to custom
  scale_fill_manual(values = c(
    "Reebok" = "steelblue",
    "Vans" = "purple"
  )) +
# change theme from default
  theme_classic() +

  # remove legend
  theme(
    legend.position = "none",
    text = element_text(size = 12)
  )
```

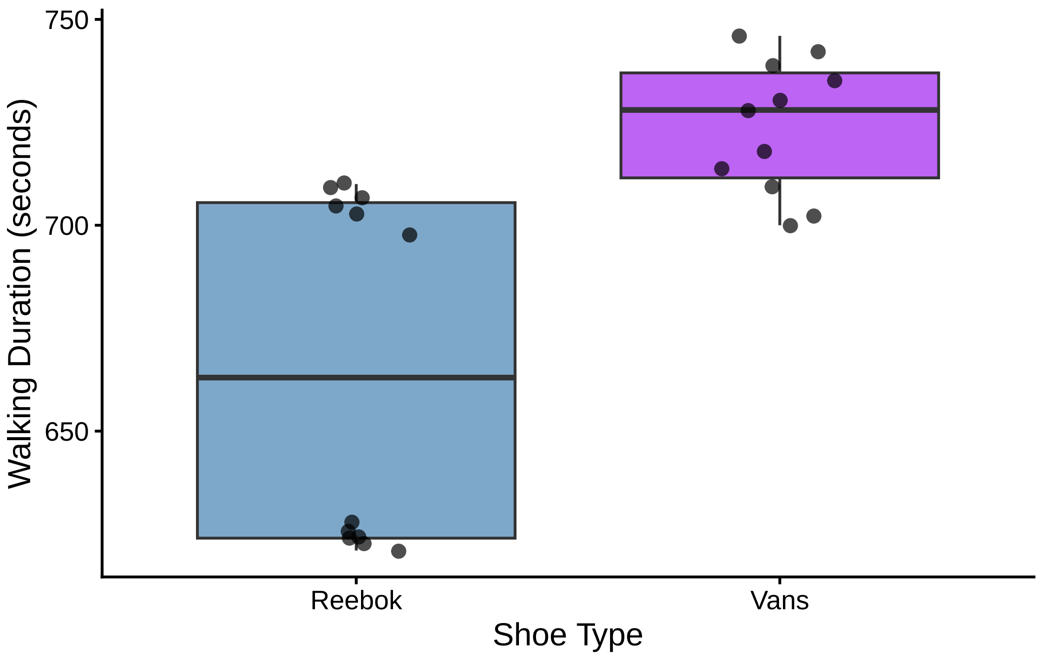


Figure 3. Walking duration by shoe type. Boxplots and jittered data points comparing walking duration (seconds) between Reebok and Vans shoes. The boxplots show the median, interquartile range, and overall spread of walking durations, while the jittered points represent individual walks. Reebok shoes generally had shorter walking durations and less variation than Vans shoes, suggesting that shoe type may influence commute time.

f. Walking duration differed between shoe types, with walks in Reebok shoes taking less time on average (664.8 s) than walks in Vans shoes (723.9 s), a difference of about 59 seconds. A Welch two-sample t-test showed that this difference was statistically significant ($t = -4.48$, $df = 14.38$, $p < 0.001$). The effect size was large (Cohen's $d = -1.81$), suggesting that shoe type had a strong effect on walking duration.